

Modeling Publication Era in English Novels Using Linguistic and Psychological Features

Background and Data

Over the past two centuries, novels have changed; this is a given. The more interesting question lies in the nature of this difference, particularly regarding linguistic features (such as the percentage of words that fall into a certain grammatical category) and psychological features (such as their tone or authenticity). This study seeks to answer the following research question:

What linguistic and psychological features are associated with whether novels were published before or after 1950, and what do these associations suggest about differences in language use between the two eras?

In the following analysis, a dataset that includes 694 English-language novels and records information regarding 107 different variables will be explored (Luoto 2021a). 93 of these variables come from the Linguistic Inquiry and Word Count (LIWC) version 2015, an analysis tool which quantifies a text's psychological and emotional aspects by breaking up the writing into tokens (individual words or punctuation markings).

The variable which most closely resembles a response variable for this research question is *PublicationYear*. All but two of the novels (1688 and 1689) were published between 1800 and 2018, so these two data points were omitted. This response was then converted into a binary variable *Post1950*, which was equal to one if the novel was released in 1950 or later and zero otherwise. This was done to simplify the modeling task into a classification problem, allowing the direct assessment of whether linguistic and psychological features can distinguish between novels of the two eras, rather than modeling gradual change over time. Furthermore, *Nationality*, which was originally broken up into American, British and then many other small-count categories, was simplified into the variable *Nationality2* which included American, British and Other.

Despite having over 100 potential predictors to choose from, most variables were immediately removed for one or more of the following reasons:

- Are too closely related to the response variable and would yield uninteresting results (e.g. the year the author was born)
- Are useless or made redundant due to the creation of *Post1950* and *Nationality2* (*PublicationYear* and *Nationality*)
- Are poorly documented or too vague to make proper conclusions about
- Are a subset of another variable (e.g. the percentages of commas, periods, etc. were removed in favor of a predictor which represented all punctuation)
- Are highly correlated with a variable of higher interest (e.g. *PositiveEmotion* and *NegativeEmotion* were relevant in the calculation of *Tone*, but *Tone* is a direct psychological feature of interest, so they were removed)

This leaves eighteen potential predictors, which are summarized in the [following section](#).

Variables

There are four LIWC summary measures which are standardized scores between 0 and 100 (Luoto 2021b). The main idea behind each algorithm will be briefly summarized and interpretations for each score will be shared below (LIWC n.d.).

1. *AnalyticalThinking*: This score is obtained based on the use of multiple types of function words. A higher number represents very logical and hierarchical thinking while a lower score typically represents more friendly or personal thinking.
2. *Clout*: This algorithm analyzes linguistic markers linked to social status and confidence and was created based on many studies done regarding people interacting with each other. A higher score represents a more confident text while a lower score represents one which is more tentative.
3. *Authentic*: This algorithm analyzes patterns in a text associated with spontaneous versus self-monitored language. A higher score represents a more honest text while a lower score represents one which is more guarded.
4. *Tone*: This algorithm integrates positive and negative emotion word frequencies into a single score. A higher number represents a more positive tone while lower numbers represent text that is more negative.

There are nine LIWC variables which measure the percentage of tokens which fall into that category.

1. *Pronoun*: The percentage of word tokens that are pronouns
2. *Article*: The percentage of word tokens that are articles
3. *Preposition*: The percentage of word tokens that are prepositions
4. *Conjunction*: The percentage of word tokens that are conjunctions
5. *Negation*: The percentage of word tokens that are negation words
6. *Verb*: The percentage of word tokens that are verbs
7. *Adjective*: The percentage of word tokens that are adjectives
8. *Number*: The percentage of “word” tokens that are numbers
9. *AllPunc*: The percentage of all tokens that are punctuation markings of any kind

There are three general descriptor variables regarding the text.

1. *WordCount*: The total number of word tokens
2. *WordsPerSentence*: The average number of words per sentence
3. *SixLetters*: The percentage of words that have more than six letters

Lastly, there are two variables remaining that represent metadata related to the novel.

1. *Sex*: Binary variable where 1 represents a male author and 2 represents a female author
2. *Nationality*: Explained in the [previous](#) section

Data Analysis

First and foremost, due to the nature of the response (binary), a binomial generalized linear model (GLM) with the logit link function was chosen. All analysis for this study was done using R.

Exploratory Data Analysis

Before fitting an initial model, exploratory data analysis was conducted to get a general understanding of the relationship between the potential predictors and the binary response.

For each of the 16 numeric predictors, side-by-side boxplots were created, one representing the variable's distribution for novels published before 1950 and the other for novels published in 1950 or later. These plots were analyzed to determine if there appeared to be a noticeable difference between the two groups (Figure 1 and Figure 2).

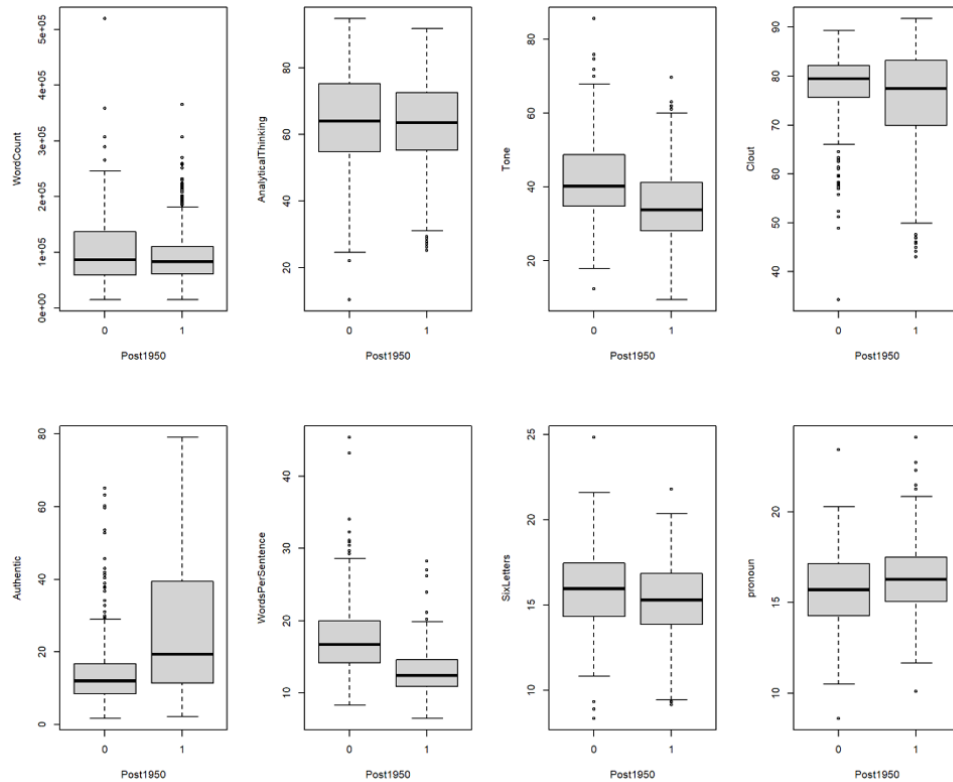


Figure 1. Boxplots 1-8

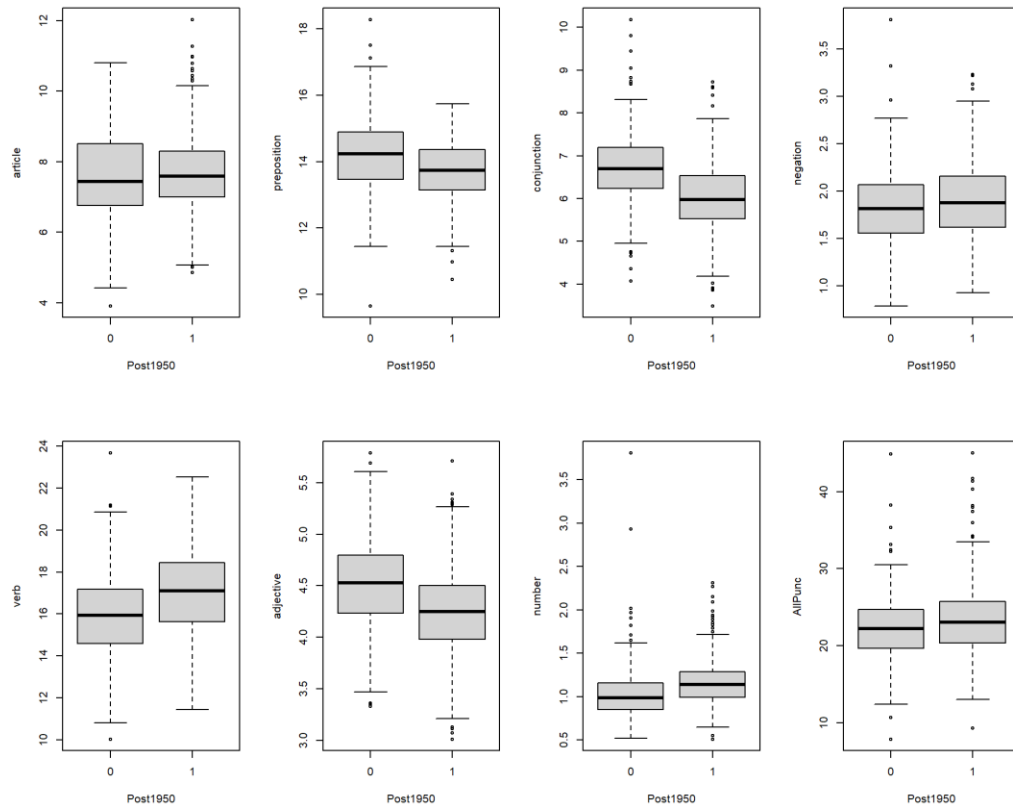


Figure 2. Boxplots 9-16

Based on the boxplots, *WordCount*, *Tone*, *Clout*, *Authentic*, *WordsPerSentence*, *Conjunction*, *Verb*, and *Adjective* all appear to differ quite a bit in pre-1950 novels and post-1950 novels.

For the two qualitative variables, *Sex* and *Nationality2*, mosaic plots were formed to assess whether they differed between novels published before and after 1950 (Figure 3).

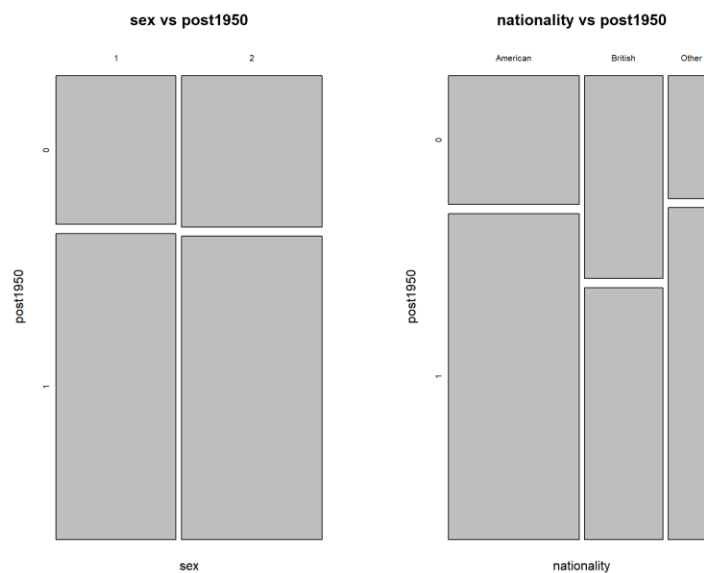


Figure 3. Mosaic Plots

For *Sex*, the proportion of male and female authors appears to be nearly the same for pre-1950 and post-1950 novels. However, the proportion of British novels which are post-1950 is much lower than that of American novels.

Lastly, to assess potential nonlinearity in the relationship between numeric predictors, eight empirical logit plots were created, one for each of the numeric predictors which displayed a noticeable pattern in their side-by-side boxplots (Figure 4). Curvature in these plots indicates a nonlinear trend.

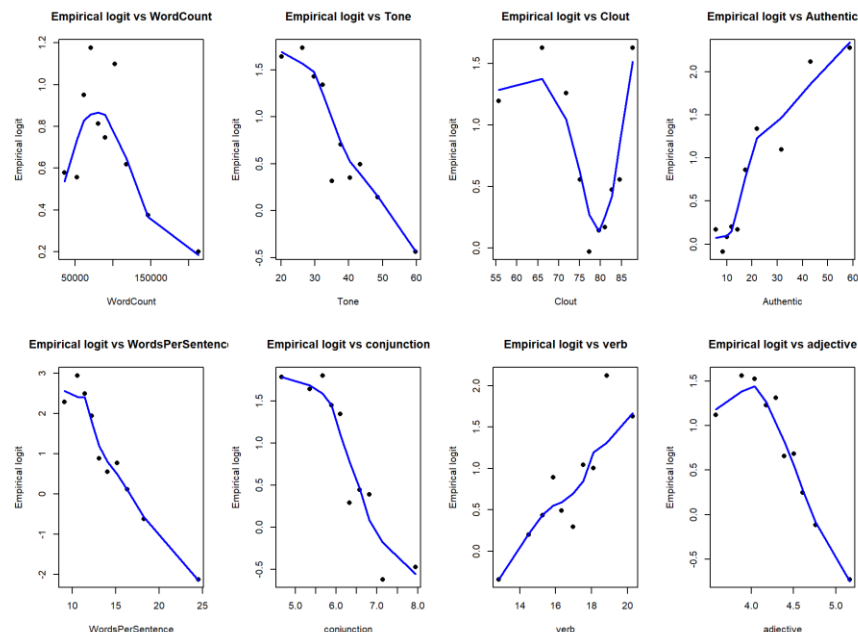


Figure 4. Empirical Logit Plots

Above, *WordCount*, *Clout*, *Conjunction* and *Adjective* all showcase curvature in their respective empirical logit plots. Therefore, this suggests the relationships between each of these numeric predictors and the outcome are nonlinear. The other four variables show relative linearity in their graphs, so there is not much concern regarding their linearity.

Initial Model Fitting and Selection

The [above](#) exploratory data analysis, as well as theoretical plausibility, will be essential in the fitting of a generalized linear model. Before fitting any models, it is important to note that the null deviance is 876.39. How much lower each model's residual deviance is in comparison is a good indication of how strong that model is compared to the null model with no predictors.

The process of model building and comparison is not simple. For that reason, Table 1 summarizes each of the main models that were fit, including their effects included, their AIC, BIC and residual deviance, and how or why they were fit.

Model name	Relative predictors in comparison to previous model	AIC	BIC	Residual Deviance	How/why they were fit
Null Model	1	878.39	882.93	876.39	For comparison to other models
Initial Model	+WordCount + Tone + Clout + Authentic + WordsPerSentence + Conjunction + Verb + Adjective + Nationality2	566.51	616.42	544.51	Predictors chosen from exploratory data analysis
Main Step Model	+ AllPunc + Number + SixLetters - WordCount - Adjective	510.46	564.92	486.46	Step() function (AIC-based) in R used to determine which main effects of our 18 should be included, using initial model as baseline
Main Step Model Adjusted	- Conjunction	510.90	560.82	488.90	Removing conjunction due to insignificant Wald test and justified using LRT
Interaction Model	+ Tone:Authentic + Tone:Clout + Tone:Nationality2 + Clout:Authentic	506.23	578.84	474.23	Adding theoretically plausible interaction terms involving main effects in model above
Interaction Step Model	- Clout:Authentic	505.00	573.07	475.00	Step() function in R used with lower scope of Main Step Model Adjusted and upper scope of Interaction Step Model
Quadratic Model	+ WordCount ² + Clout ² + Adjective ² + Conjunction ²	507.47	593.70	469.47	Adding the four quadratic effects of variables which showed concern regarding linearity in EDA
Quadratic Step Model	- WordCount ² - Clout ²	503.72	580.87	469.72	Step() function used with same logic as the Interaction Step Model

Table 1. Model Selection Process

Throughout this process, likelihood ratio tests were utilized to formally compare (nested) models using deviance. When the resulting p-value was less than 0.05, it was determined that the more complicated model was worth its additional complexity. This method was first used to compare the Main Step Model with the simpler Main Step Model Adjusted. The resulting p-value was 0.1185, so the removal of *Conjunction* from the model was justified.

Likelihood ratio tests were also used to compare the models with varying interaction terms to find the best selection without going outside the range of determined theoretical plausibility. The test comparing the Interaction Step Model (some interactions) with the more complicated Interaction Model (all plausible interactions) yielding a p-value of 0.3805, suggesting the *Clout:Authentic* interaction should not be included in the model. On the other hand, the test comparing the Main Step Model Adjusted (no interactions) with the Interaction Step Model (some interactions) output a p-value of 0.0076, suggesting that the three interactions included should remain.

The same process was done to compare the models with varying quadratic terms. Both tests yielded p-values greater than 0.05, suggesting that no quadratic terms should remain in the model. The last change that was made before declaring a final GLM was the removal of *number* from the model. This was due to its unusually large coefficient (exponentiated).

Interpreting the Results of the GLM

First and foremost, to interpret the resulting coefficients and confidence intervals from the final GLM, they need to be exponentiated. This allows them to be interpreted as odds rather than log-odds. Four main effects are clearly able to be interpreted based on their respective 95% confidence intervals due to the inclusion of no interaction effects with them. None of these four intervals include one, which signifies that each of these effects are significant at a 95% confidence level.

WordsPerSentence (0.57 to 0.69): Each one-word increase per sentence decreases the odds of a novel being published in or after 1950 by about 31-43%. This is a stable negative relationship.

Verb (1.16 to 1.61): Each one percent increase in verb usage increases the odds of a novel being published in or after 1950 by 16-61%. This is a stable positive relationship

AllPunc (0.77 to 0.88): Each one percent increase in punctuation decreases the odds of a novel being published in or after 1950 by 12-23%. This is a stable negative relationship.

SixLetters (1.29 to 1.72): Each one percent increase in words longer than 6 letters increases the odds of a novel being published in or after 1950 by 29-72%. This is a stable positive relationship.

As for the main effects which also have interaction effects in the model, interpretation is a little bit trickier. The 95% confidence interval (exponentiated) for *tone* is (0.30 to 0.81). This means that holding other variables at reference levels for interactions, a one-unit increase in *Tone* score is associated with a 19-70% decrease in the odds of a novel being published in or after 1950. This isn't quite sufficient since *Authentic* and *Clout* scores can not be equal to zero; however, combined with the side-by-side boxplots, it is safe to say that pre-1950 novels appear to generally have more positive tones (higher *Tone* scores). The confidence intervals for *Clout* and *Authentic* scores both include one, so their main effects cannot be considered significant.

Lastly, each of the four interaction effects in the generalized linear model have the following 95% confidence intervals:

Tone:Authentic (1.002 to 1.009): As *authentic* increases, the effect of *Tone* on the odds of a novel being 1950 or later becomes slightly less negative

Tone:Clout (1.002 to 1.013): As *Clout* increases, the effect of *Tone* on the odds of a novel being 1950 or later becomes slightly less negative.

Tone:Nationality2British (0.9992 to 1.1079) and *Tone:Nationality2Other* (0.8397 to 1.0053) both include one, so these effects can't be considered significant.

Diagnostics

In diagnostics, three criteria were evaluated: model fit and classification performance, model misspecifications, and influential points.

To assess model fit, the first step taken was to plot a standardized quantile residual versus linear predictor plot, as seen in Figure 5. Due to there being no clear patterns in the plot, there is no cause for immediate concern.

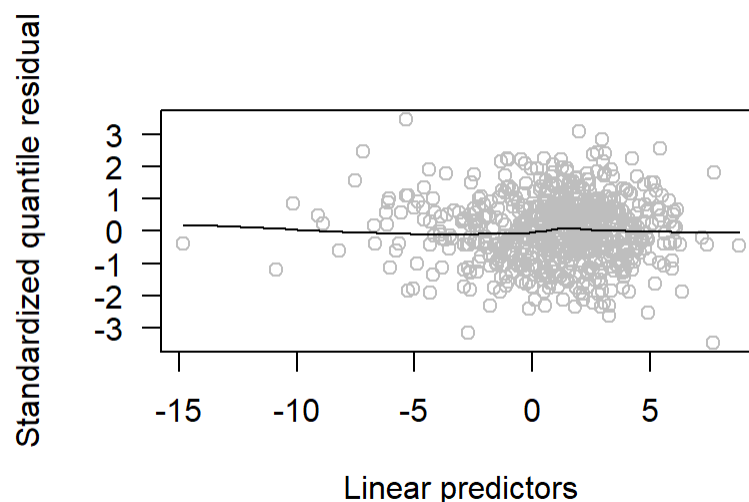


Figure 5. Quantile Residual vs. Linear Predictor Plot (GLM)

Additionally, a ROC (Receiver Operating Characteristic) curve (which compares the false positive rate with the true positive rate), was fit (Figure 6). Since the curve is very close to the top-left corner, it can be said that the model does a good job separating pre-1950 and post-1950 novels. This is also supported by a large area under the curve (AUC) of 0.9068.

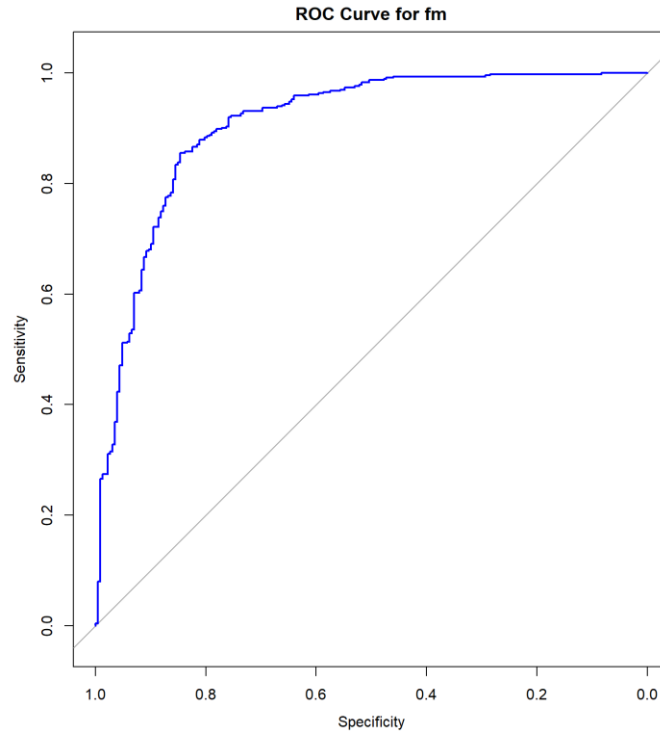


Figure 6. ROC Curve (GLM)

Next, to assess the potential of a link function misspecification, a working response vs linear predictor graph was created with a Lowess line (Figure 7). There is a tiny bit of curvature in the center, which means the link function could be slightly misspecified. However, overall, it is quite linear, suggesting that it is not a huge concern.

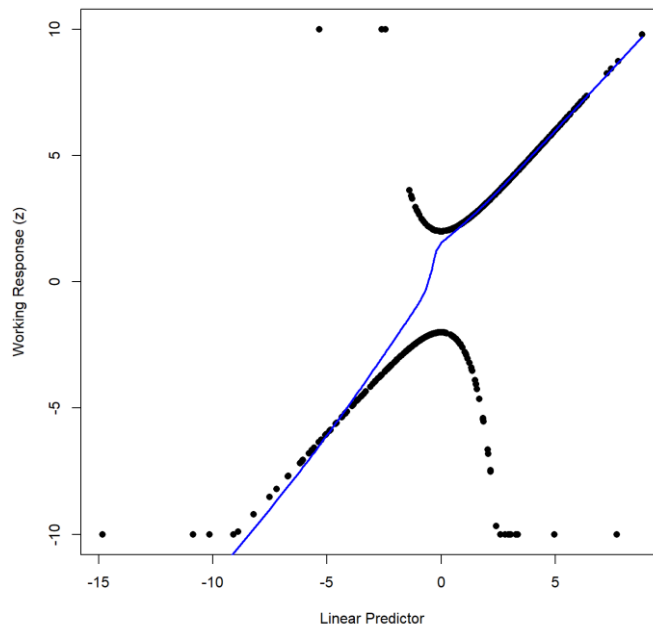


Figure 7. Working Response vs. Linear Predictor Plot (GLM)

Furthermore, to assess linearity of numeric predictors, quantile residual plots were created for each one in the final GLM (Figure 8). All plots show no clear patterns, suggesting that nonlinearity is not an issue. However, it is clear from the plots that potential outliers are a concern.

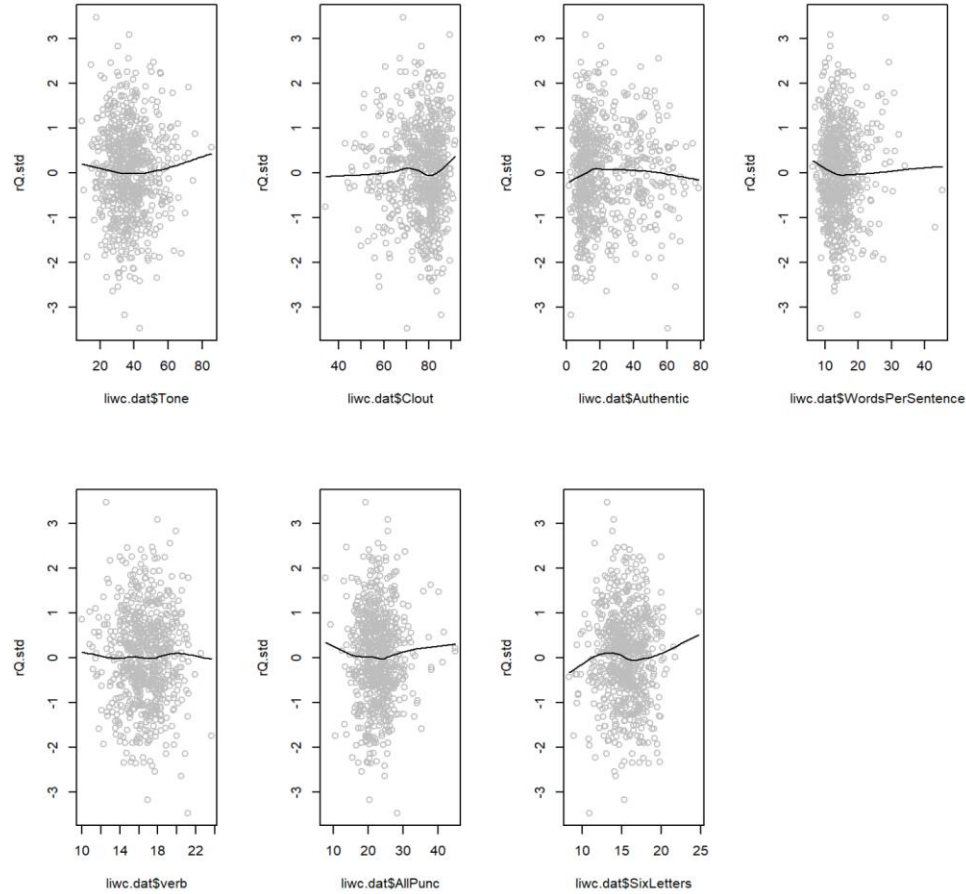


Figure 8. Quantile Residual Plots for Each Numeric Predictor (GLM)

So, the last step taken in diagnostics was to fit a model without the influential points in it to see if the model changed in a significant way. Influential points were distinguished using DFFITS criteria with a typical cutoff of $2 \times \sqrt{p/n}$, where n is the number of observed novels in the dataset and p is the number of parameters in the model. Once these influential points were identified using this criterion, they were removed from the dataset, and a new model was created.

Many of the resulting model's coefficients differed quite a bit, making exact interpretations a bit different. However, when just looking at general patterns, both models were the same.

WordsPerSentence and *AllPunc* had stable negative relationships while *Verb* and *SixLetters* had stable positive relationships. Additionally, the main effect of *Tone* was still negative, and the impact *Authentic* and *Clout* scores had on this effect were still very similar. For this reason, the influential points are not deemed significant enough to justify their potential removal or further investigation.

Mixed Model

Although the GLM above appears to be a good fit for the data and provides a ton of valuable information regarding the research question, there is one potential issue due to the nature of the dataset. Specifically, many authors contribute multiple novels included in the data, but most appear exclusively in either the pre-1950 or post-1950 group. This trend is displayed in Figure 9 below. As a result of this, the GLM can inflate the fixed effects and confounds author style with the time period.

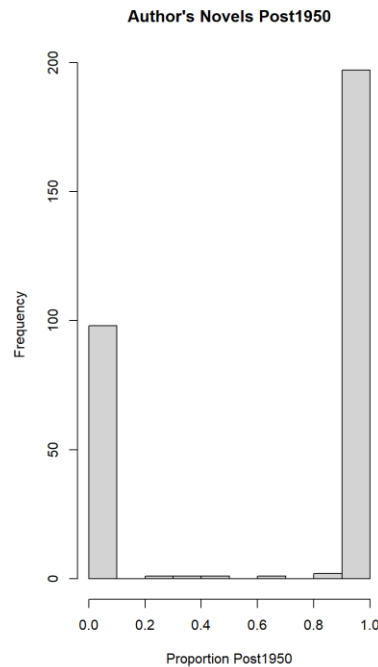


Figure 9. Distribution of Authors' Proportion of Novels Published Post-1950

To account for this structure, a generalized linear mixed model (GLMM) was fit with a random intercept for *AuthorID*. This model accounts for author-to-author differences in writing style with goal of assessing whether the relationships observed from the GLM remain robust after accounting for author-level clustering. So, although this model will provide a more conservative estimate of effects by adjusting for author-level clustering, it does not eliminate all confounding in the structure of the dataset.

The same set of predictors and interaction terms identified in the final GLM model were included in the GLMM. The large variance (268.8) associated with *AuthorID* reinforces the importance of accounting for author-level variation in the model. The results showed that, after accounting for author-level variation, the main effects *Tone*, *WordsPerSentence*, *AllPunc*, *SixLetters* and *Verb* all remained significant. This suggests they are robustly associated with publication era and are not solely driven by differences between authors' writing styles.

On the other hand, the interaction effects between *Tone* and both *Authentic* and *Clout*, which were significant in the GLM, did not remain significant in the mixed model. This suggests these interactions were likely driven by differences in which authors are represented in each period.

Discussion

Association

This study investigated whether linguistic and psychological features of English novels were associated with whether they were published before or after 1950. Overall, the results suggest that linguistic features do contain strong indications of publication era, with the strongest coming from structural aspects. However, the exact interpretation of these effects depends on accounting for author-level variation in the dataset.

Linguistic Features

Across both the GLM and GLMM, several linguistic features consistently distinguished between pre-1950 and post-1950 novels. Novels published after 1950 tend to have shorter sentences, lower punctuation usage, a higher proportion of verbs, and a higher proportion of longer words. These patterns may reflect a broader shift in English writing style, with more modern texts tending to use more verbs and shorter, less punctuation-heavy sentences. Importantly, these effects remained stable after introducing a random intercept for AuthorID.

These results align with prior research which looked at the variation of sentence length and concluded that sentences have generally become shorter over time (Rudnicka 2018). Overall, they suggest that there are stable linguistic shifts in English novels across the 1950 boundary.

Psychological Features

In terms of psychological features, the results were more mixed. *Tone* was a significant predictor in both frameworks, indicating that novels published before 1950 tend to have more positive tones. This conclusion is largely consistent with a study on the expression of emotions through the 20th century (Acerbi 2013). The analysis suggested a possible difference between American and British novels in how tone varied across publication eras, although this interaction was not statistically significant. This distinction between American and British English novels could make for an interesting research question in a future study.

Other measures such as the *AnalyticalThinking*, *Clout*, and *Authentic* scores did not display such stable effects, particularly (for the latter two) once author-level variation was accounted for. This indicates that psychological features like these are more sensitive to author-to-author variation than to publication era.

Preferred Model

A key result of this study is the difference between the GLM and GLMM frameworks. In general, the inclusion of a random intercept for AuthorID shifts the interpretation of fixed effects to within-author comparisons. However, because most authors' novels are observed in only one era (pre-1950 or post-1950), the model has very limited information regarding within-author comparisons. Therefore, the random effect primarily serves to adjust for dependence among observations from the same author and to account for systematic differences in authors' baseline writing style.

Because the GLM assumes independence of observations, it is not fully appropriate for this dataset. Ignoring this dependence can lead to underestimated standard errors and inflated significance, meaning that the GLM may overstate the strength of relationships between

predictors and publication era. The observation that most effects in the GLMM are lower in significance, a couple of which even become insignificant, is strong evidence of this effect. Although the model has clear limitations, particularly in its restricted ability to fully separate time and author effects due to limitations in the data, it is selected as the preferred model.

Limitations

Several limitations should be noted regarding this analysis and its interpretations. Most importantly, as emphasized [above](#), due to the nature of the data used, there is strong confounding between author and publication era, as most authors contribute texts to only one of the categories. The result is that, even with the fitted GLMM, this structure can only be partially accounted for. Therefore, it is not possible to fully distinguish whether observed effects are driven by changes over time or differences between authors. This limits the ability to make strong interpretations about how language has evolved over time.

In addition, the binary simplification of publication year is another potential limitation. The nature of this response could obscure gradual linguistic change. Furthermore, the lack of documentation for many variables in this dataset, as well as the noticeable absence of any information regarding the genre of each novel, are concerns.

Lastly, the nature of how LIWC variables are calculated also needs to be acknowledged. For example, the *Tone* score of each novel is calculated based on an algorithm which defines words which invoke either positive emotion or negative emotion. There is certainly a level of subjectivity involved in this calculation that needs to be recognized when making conclusions regarding the difference in novels' tones before and after 1950. This highlights the difficulty in analyzing literature, particularly with regards to its psychological aspects.

Future Improvements and Final Words

In future studies, the confounding limitation could be addressed by finding or collecting data from authors who write across long time spans. Modeling publication year as a continuous variable rather than a binary outcome would also allow for the detection of more gradual linguistic change. Additionally, fine-tuning the data collection process to better suit the needs of this research question could vastly improve the results of a future study.

Overall, the GLMM provides the most reliable framework for this analysis, indicating that linguistic features, such as sentence length, punctuation usage, verb frequency and word length, are consistently associated with publication era. These relationships remain robust after accounting for author-level variation, suggesting that they are not solely explained by differences in authors' writing styles. In contrast, psychological features provide less stable association, suggesting that these effects are more indicative of differences between authors than between pre- and post-1950 novels.

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